

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2013

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

- 4.** Which one of the following components of blood is involved in the clotting after injury?
- A. Plasma
 - B. White blood cells
 - C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2013.

1. Heat is transferred through a vacuum by
 - A. Convection and conduction
 - B. Convection only
 - C. Conduction and radiation
 - D. Radiation
2. The **best** procedure of separating a mixture of sand, maize and salt is
 - A. Picking, dissolving, filtering, evaporation
 - B. Dissolving, filtering, picking, evaporation
 - C. Picking, filtering, dissolving, evaporation
 - D. Dissolving, decanting, picking, evaporation
3. To stop a moving object, the force applied should be
 - A. In the same direction as that of the moving object
 - B. Opposite and less than that of the moving object
 - C. Equal and the opposite direction to that of the moving object
 - D. Perpendicular and more than that of the moving object
4. The set up shown below can be used to investigate the presence of a certain component of soil.

The soil component investigate is

- A. Organic matter
 - B. Water
 - C. Air
 - D. Mineral
5. Which of the following pairs of processes occurs when temperature decreases?
 - A. Melting and condensation
 - B. Freezing and melting
 - C. Contraction and evaporation
 - D. Freezing and condensation
6. Which one of the following animals has scales on its body?
 - A. Chameleon
 - B. Whale
 - C. Toad
 - D. Newt
7. The main reasons why mothers are encouraged to breast feed their infants is because breast milk
 - A. Is free from disease causing germs
 - B. Is at the right temperature for infants
 - C. Boosts infant immunity
 - D. Is easily digested by infants
8. Splash erosion can be prevented by
 - A. Use of gabions
 - B. Mulching
 - C. Terracing

- D. Contour farming
9. Newton is the unit for measuring
- A. Weight
 - B. Mass
 - C. Volume
 - D. Pressure
10. Soil that is good for modeling has
- A. High water drainage
 - B. Rough capillarity
 - C. High capillarity
 - D. Large air space
11. The following are processes that take place during birth in humans
- i) Dilation of the cervix
 - ii) Construction of the uterus
 - iii) Bursting of the amniotic bag
 - iv) Pushing out of the baby
 - v) Cutting of the umbilical cord
- Which of the following is the correct order of the processes?
- A. (i) (iii) (ii) (iv) (v)
 - B. (i) (iii) (iv) (ii) (v)
 - C. (ii) (i) (iii) (iv) (v)
 - D. (ii) (iv) (iii) (i) (v)
12. A farmer practicing stall feeding is likely to feed animals on
- A. Pasture and fodder
 - B. Pasture and concentrates
 - C. Fodder and concentrates
 - D. Concentrates only
13. Which of the following statements is **true** above vena cava? It carries
- A. Deoxygenated blood
 - B. Blood from the lungs
 - C. Blood under high pressure
 - D. Blood from the heart
14. Which of the following materials are both opaque?
- A. Mirror and fresh milk
 - B. Paper smeared with oil and frosted glass
 - C. Fresh milk and frosted glass
 - D. Mirror and paper smeared with oil
15. Which one of the following pairs is a source of electricity?
- A. Dam and a dynamo
 - B. Wind-driven turbine and dam
 - C. Torch and a dynamo
 - D. Dry cell and a geothermal generator.

16. Which of the following statements is true about cumulus clouds?
- Their presence is sign of rain
 - Appear at low altitude and spread out in flat layers.
 - They are shaped like mountains
 - They are dark grey in colour
17. Which of the following planets are in the 2nd and 6th positions from the sun?
- Mars and Jupiter
 - Venus and Saturn
 - Earth and Saturn
 - Venus and Jupiter
18. In which of the pairs of levers is the load, effort and fulcrum likely to be in similar positions when in use?
- Crowbar and claw hammer
 - Wheelbarrow and spade
 - Crowbar and spade
 - Crow hammer and wheelbarrow
19. Which of the following pairs of objects is correctly classified into magnetic and non magnetic?
- | Magnetic | non magnetic |
|-----------------|---------------------|
| A. Glass sheet | tin can |
| B. Steel needle | pencil lead |
| C. Copper coin | steel wool |
| D. Razor blade | staple pin |
20. The diagram below represent a set that is used to demonstrate a certain aspect of matter

The aspect investigated is

- Radiation
 - Convection
 - Conduction
 - Expansion
21. The following are uses of some components of air:
- Germination
 - Preserving soft drinks
 - Electric bulbs
 - Burning
- Which two uses are for the component that makes up 21% of air
- (i) and (iii)
 - (ii) and (iii)
 - (i) and (iv)
 - (ii) and (iv)
22. A pupil was walking towards the northern direction past an air strip and noticed a wind vane pointing to her right. In which direction was the wind blowing from?
- East

- B. South
- C. North
- D. West

23. The diagram below represents a electromagnet

Which of the following energy transformations occurs in the electromagnet

- A. Electrical->magnetic ->chemical
- B. Chemical ->magnetic ->electrical
- C. Chemical-> electrical
- D. Magnetic ->chemical->electrical

24. Vaccines given at birth protect an infant against

- A. Diphtheria and tuberculosis
- B. Tuberculosis and poliomyelitis
- C. Tetanus and diphtheria
- D. Whooping cough and poliomyelitis

25. Farmers apply farm yard manure to the soil to

- A. Improve soil structure
- B. Release nutrients needed by plants immediately
- C. Provided specific nutrients
- D. Provided nutrients in right amounts

26. Which pair of plants stores food in the roots?

- A. Sweet potato and carrot
- B. Irish potato and arrow root
- C. Carrot and irish potato
- D. Onions and cassava

27. Which of the following groups of food crops is **correctly** classified into cereals ,legumes and fruits?

Cereals	legumes	fruits
A. Peas	millet	orange
B. Apple	green grams	beans
C. Rice	ground nuts	pawpaw
D. Wheat	bananas	sorghum

28. The diagram below shows some parts flower.

The parts representing the stigma and filaments are

- A. K and L
- B. L and N
- C. K and M
- D. N and M

29. Which of the following food chains correct?

- A. Grass ->grasshopper ->goat
- B. Maize->mice ->snake

- C. Sunflower ->butterfly
 - D. Grass ->antelope->rhino
30. Which of the following groups of animals feeds mainly provides a diet of proteins, carbohydrates and fats respectively?
- A. Desmodium, clover, sweet potatoes
 - B. Lucern, maize stalk, oats
 - C. Napier grass, sweet potato vines, maize
 - D. Lucern, desmodium, napier grass.
31. Which of the following groups of foods would provide nutrients suitable for a child suffering from kwashiorkor?
- A. Sorghum, carrots, sweet potatoes
 - B. Oranges, irish potatoes, sugar cane
 - C. Beef ,peas, milk
 - D. Mangoes, arrowroots, spinach
32. Bilharzias can be prevented by
- A. Boiling water before drinking
 - B. Wearing gumboots when walking in stagnate water
 - C. Draining stagnate water from around the homestead
 - D. Covering latrines after use
33. Which of the following uses of water are recreational only
- A. Transport, cleaning, swimming
 - B. Cleaning, boat racing, mixing chemicals
 - C. Swimming, fishing, boat racing
 - D. Fishing, transport, mixing chemicals
34. Which of the following statements about HIV infected persons is **correct** :
- A.** Can only be taken care of health workers
 - B.** Need to visit hospital for core
 - C.** Should be provided with a balance diet
 - D.** Should be discouraged from playing with others
35. Excessive uses of herbicides and pesticides affect the soil by reducing the
- A. Number of animals in the soil
 - B. Nutrients in the soil
 - C. Humus content in the soil
 - D. Water retention in the soil
36. Which of the following parts of the human breathing system are both involved in cleaning, warming and moistening air.?
- A. Diaphragm and trachea
 - B. Lungs and diaphragm
 - C. Nose and lungs
 - D. Trachea and nose.
37. Which of the following pairs of living stock parasite control methods liver flukes?
- A. Rotational grazing and deworming

- B. Dipping and spraying of animals
- C. Rotational grazing and spraying of animals
- D. Deworming and dipping

38. In which of the following are characteristics of solids and liquids **correctly** matched?

Solid	liquids
A. Definite volume	definite shape
B. No definite volume	definite shape
C. Definite volume	no definite shape
D. No definite shape	definite volume

39. Which of the following objects consists only object that sink in water?

- A. Piece of chalk, steel needle, cork
- B. Nail, piece of glass, stone
- C. Bottle top, nail, cork
- D. Piece of chalk, plastic, bottle, polythene bag

40. The following are reasons for fitting a tap near the bottom of a tank except?

- A. Making gravitational force to act on water
- B. Making water flow faster
- C. To increase pressure
- D. To let all water out

41. The main reason why murram is placed on road is to

- A. ABSORB RAIN WATER
- B. Make the road last longer
- C. Increase grip with tyres
- D. Make the road attractive

42. Which of the following groups of products are correctly classified according to the farm animals producing them

CATTLE	SHEEP	GOAT
A. Dairy	beef	wool
B. Beef	dairy	wool
C. Mutton	wool	dairy
D. Beef	mutton	dairy

43. The following processes are involved in the formation of zygote in plants:

- i) Growth of pollen tube
- ii) Pollination
- iii) Fusion
- iv) Pollen tube breaks

The correct order of processes is

- A. (ii), (i), (iv), (iii)
- B. (i), (ii), (iv), (iii)

- C. (ii), (i), (iii), (iv)
D. (iv), (i), (iii), (ii)
44. Which of the following only consists of energy giving foods?
A. Lemon and honey
B. Butter and coconut
C. Eggs and fish
D. Bread and pineapple
45. Which of the following statements describe soft water? Soft water
A. Is clean and clear
B. Requires a lot soap for washing
C. Is safe for drinking
D. Is water that has been boiled
46. In an activity to investigate mixing of liquid, liquid mixed with liquid Q, liquid R did not mixed with liquid P and Q. liquid R mixed with cooking oil. Which one of the following correctly represents liquid P,Q, and R?
- | P | Q | R |
|-------------|----------|----------|
| A. Water | milk | kerosene |
| B. Kerosene | water | milk |
| C. Milk | kerosene | water |
| D. Water | kerosene | milk |
47. The diagram below represents a set up used to demonstrate a certain property of light.
- Which none of the following statements is correctly about the set up? light is
- A. Reflected from P to the eye
B. Reflected from Q to the eye
C. Reflected from Q to the eye
D. Reflected from P to the eye
48. Which one of the following pairs of materials will pollute soil
A. Saw dust and artificial fertilizer
B. Plastic and artificial fertilizer
C. Wood ash and saw dust
D. Plastics and iron fillings
49. In a certain investigation pupils used a straight pipe to observe a candle flame. They then bent the pipe and observed again.
Which aspect of light was being investigated?
A. Formation of shadows
B. Uses of light
C. Sources of light
D. How light travels
50. A pupil was blindfolded and asked to listen as a bell was ring from different positions.
Which aspect of sound was being investigated
A. Direction of sound

- B. Effects of sound pollution
- C. Loud and soft sounds
- D. Meaning of special sounds